Omics Research Pipeline: Top Biases by Stage

Top 3 most-cited biases per pipeline stage | 1,978 citations across 145 biases in 7 omics categories

1. Data Production

468 citations (23.7%)

- Sample Heterogeneity (26)
- Sample Variability (23)
- Dropout / Sparsity (22)

2. Technical/Instrumental

456 citations (23.1%)

- Batch Effects (27)
- Platform Variability (24)
- Short-read Limitations (21)

3. Computational/Analytical

492 citations (24.9%)

- Batch Correction (25)
- Pipeline Heterogeneity (23)
- Data Integration (21)

4. Reporting/Interpretation

164 citations (8.3%)

- Black-Box Models (18)
- Methods Transparency (17)
- Publication Bias (17)

5. Other Challenges

398 citations (20.1%)

- Standardization Gaps (22)
- Privacy & Regulation (21)
- Benchmark Gaps (20)

Rank within stage
1st (most cited)
2nd
3rd

0 5 10 15 20 25

Number of Citations